National Exit Exam Questions (1-96)

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1. What is the primary key in a relational database?

A. A unique identifier for each row in a table.

B. A foreign key from another table.

C. A key used for encryption.

D. An index on a non-unique column.

Answer: A. A unique identifier for each row in a table.

Explanation: In a relational database, the primary key is a column or set of columns that uniquely identifies each row in a table. It ensures that no two rows can have the same value for the primary key, enforcing uniqueness and integrity.

2. The E-R data model is based on a perception of the real world that consists of a set of basic objects called ______

A. Objects

B. Classes

C. Entities

D. Attributes

Answer: C. Entities

Explanation: In the Entity-Relationship (E-R) data model, the real world is modeled as a set of basic objects called entities, which represent real-world objects or concepts. Entities have attributes that define their properties, and relationships can be established between entities.

3. Which scheduling algorithm is commonly associated with the concept of aging, where the priority of a task increases the longer it waits in the ready queue?

A. Round Robin (RR)

B. Shortest Job Next (SJN)

C. Priority Scheduling

D. Shortest Remaining Time First (SRTF)

Answer: C. Priority Scheduling

Explanation: In Priority Scheduling, the concept of aging is used to prevent starvation of lower-priority tasks. Aging gradually increases the priority of a process the longer it waits in the queue, ensuring that all tasks eventually get executed.

4. In computers, subtraction is carried out generally by __________

A. 1‘s complement method

B. 2‘s complement method

C. Signed magnitude method

D. BCD subtraction method

Answer: B. 2’s complement method

Explanation: In computers, subtraction is commonly performed using the 2's complement method. This method simplifies subtraction by converting it into addition, where the subtrahend is negated using its 2's complement and then added to the minuend. This approach avoids the need for complex subtraction circuits.

5. Which reasoning technique in artificial intelligence is based on a set of rules and logical inference to derive conclusions?

A. Abductive reasoning

B. Analogical reasoning

C. Deductive reasoning

D. Inductive reasoning

Answer: C. Deductive reasoning

Explanation: Deductive reasoning in artificial intelligence is based on applying a set of predefined rules and logical inference to draw conclusions from given premises. It moves from general principles to specific conclusions, making it a key method in rule-based AI systems.

6. What is the purpose of normalization in database design?

A. To reduce redundancy and improve data integrity

B. To speed up query performance

C. To create complex relationships between tables

D. To encrypt sensitive data

Answer: A. To reduce redundancy and improve data integrity

Explanation: Normalization in database design is the process of organizing data to reduce redundancy and dependency. This is achieved by dividing large tables into smaller, related tables and ensuring that relationships between the tables are well-defined. This helps improve data integrity and consistency.

7. If a problem can be solved by combining optimal solutions to non-overlapping problems, the strategy is called _______.

A. Dynamic programming

B. Divide and Conquer

C. Recursion

D. Greedy

Answer: B. Divide and Conquer

Explanation: Divide and Conquer is a strategy that solves a problem by breaking it down into smaller, non-overlapping subproblems, solving each independently, and then combining their solutions to solve the original problem. Examples include merge sort and quicksort.

8. Which attribute of <img> tag is used when we want to display a text in place of an image when, due to any reason, the image is not displayed?

A. rep

B. pla

C. alt

D. im

Answer: C. alt

Explanation: The alt attribute of the <img> tag is used to specify alternative text that will be displayed if the image cannot be loaded. This is important for accessibility and helps screen readers describe the content of the image to users with visual impairments.

9. Which of the following concepts in OOP allows a class to inherit attributes and methods from another class?

A. Inheritance

B. Abstraction

C. Encapsulation

D. Polymorphism

Answer: A. Inheritance

Explanation: Inheritance in Object-Oriented Programming (OOP) allows a class (called a subclass or derived class) to inherit attributes and methods from another class (called a superclass or base class). This enables code reusability and hierarchical relationships between classes.

10. Which of the following is the top-level object?

A. Root object

B. Document object

C. JavaScript object

D. Window object

Answer: D. Window object

Explanation: In web browsers, the Window object is considered the top-level object. It represents the browser window and serves as the global object in the browser environment. All other objects, such as the Document object, are properties or methods of the Window object.

11. Which function of modern operating systems involves managing hardware resources such as CPU, memory, and I/O devices?

A. File Management

B. Device Management

C. Process Management

D. Security Management

Answer: C. Process Management

Explanation: Process management is a core function of modern operating systems, responsible for managing hardware resources such as the CPU, memory, and I/O devices. It ensures that processes (programs in execution) have access to the necessary resources and manages tasks like process scheduling, multitasking, and memory allocation.

12. In which phase of the compilation process does the source program get broken into its constituent parts?

A. Semantic Analysis

B. Syntax Analysis

C. Synthesis

D. Lexical Analysis

Answer: D. Lexical Analysis

Explanation: In the lexical analysis phase of the compilation process, the source program is broken into its constituent parts, known as tokens. These tokens represent basic syntactical elements such as keywords, operators, identifiers, and literals, which are used by the later phases of the compiler.

13. Which part of the computer system is responsible for converting machine processed data to different formats that are understandable by user?

A. Communication components

B. Output unit

C. Storage unit

D. Input unit

Answer: B. Output unit

Explanation: The output unit of a computer system is responsible for converting machine-processed data into a format that is understandable by the user, such as visual displays on a monitor, printed documents, or audio output.

14. What is the purpose of interfaces in Java?

A. To create multiple constructors for a class

B. To define the state and behavior of an object

C. To define a blueprint of a class

D. To achieve multiple inheritance

Answer: D. To achieve multiple inheritance

Explanation: In Java, interfaces are used to achieve multiple inheritance. While Java doesn't support multiple inheritance through classes, it allows a class to implement multiple interfaces. Interfaces provide a way to define methods that must be implemented by any class that uses them, thus allowing different behaviors to be inherited from multiple sources.

15. Which one of the following has the fastest access time and the highest cost?

A. Register

B. Magnetic disc

C. Main memory

D. Cache memory

Answer: A. Register

Explanation: Registers have the fastest access time and the highest cost compared to other types of memory. They are small storage areas located directly within the CPU, which makes accessing them extremely fast. However, due to their limited size and complex design, they are also the most expensive type of memory in a computer system.

16. Which OOP concept allows a class to implement multiple interfaces in Java?

A. Polymorphism

B. Encapsulation

C. Inheritance

D. Interface

Answer: A. Polymorphism

Explanation: Polymorphism allows a class in Java to implement multiple interfaces. It enables objects to be treated as instances of their parent interface or class, allowing different classes to define their own unique implementations of interface methods. By implementing multiple interfaces, a class can adopt behaviors defined by those interfaces, leveraging polymorphism to interact with them.

17. Which database design concept is commonly used in online gaming platforms to handle real-time interactions between players?

A. Replication

B. NoSQL databases

C. Distributed databases

D. Partitioning

Answer: C. Distributed databases

Explanation: In online gaming platforms, distributed databases are commonly used to handle real-time interactions between players. Distributed databases allow data to be stored across multiple servers, which improves scalability, reduces latency, and enhances the performance of real-time interactions in a multiplayer environment. This approach enables faster access to data and better handling of a large number of concurrent users.

18. Which one of the following is the correct syntax for document type declaration in HTML5?

A. </doctype html>

B. <!doctype! Html>

C. <doctype! Html>

D. <!doctype html>

Answer: D. <!doctype html>

Explanation: In HTML5, the document type declaration is written as <!doctype html>, which tells the browser that the document is an HTML5 document. This declaration is simple and does not require a specific DTD (Document Type Definition), unlike earlier versions of HTML.

19. Which SQL command is used to modify existing data in a table?

A. ALTER

B. UPDATE

C. MODIFY

D. CHANGE

Answer: B. UPDATE

Explanation: The UPDATE SQL command is used to modify existing data in a table. It allows you to change the values of one or more columns in one or more rows based on a specified condition.

20. What is the term for a named block of code that performs a specific task?

A. Condition

B. Variable

C. Function

D. Loop

Answer: C. Function

Explanation: A function is a named block of code designed to perform a specific task. It can be called and executed whenever needed, and it may return a value based on its operations. Functions help in organizing code, promoting reusability, and simplifying complex problems.

21. Which of the following is FALSE in Data Structures and Algorithms?

A. A tree contains a cycle

B. Tree is a non-linear data structure

C. A tree is a connected graph

D. A tree with n modes contains (n-1) edge

Answer: A. A tree contains a cycle

Explanation: A tree is an acyclic graph, meaning it does not contain any cycles. A tree is defined as a connected graph with no cycles, which is a fundamental property of trees in data structures and algorithms.

22. What is the role of a database administrator (DBA)?

A. Designing user interfaces for database applications

B. Developing front-end application

C. Writing complex SQL queries

D. Managing and maintaining databases

Answer: D. Managing and maintaining databases

Explanation: A DBA is primarily responsible for the overall management, maintenance, and security of databases. Their duties include tasks like: Ensuring the database runs efficiently and reliably, Implementing and managing backups and recovery, Monitoring performance and optimizing queries, Ensuring data security and integrity and Installing and upgrading database software.

23. Among the following which kind of algorithm is used in the Game tree to make decisions of Win/Lose?

A. Heuristic Search Algorithm

B. Min/Max Algorithm

C. Depth First Search Algorithm

D. Greedy Search Algorithm

Answer: B. Min/Max Algorithm

Explanation: The Min/Max algorithm is widely used in decision-making processes in game trees, particularly in two-player games like chess or tic-tac-toe. It helps in determining the best move by assuming that one player will maximize their score (the "max" player) while the opponent will minimize it (the "min" player). This approach allows the algorithm to simulate and evaluate all possible future moves to determine the best course of action for the current player.

24. Which one of the following is not a reason for the need of backups?

A. Security incidents

B. Upgrades and migrations

C. Hardware failures

D. Authorization

Answer: D. Authorization

Explanation: Authorization refers to controlling user access to resources in a system and is not a reason for needing backups.

25. Which command in Microsoft Windows will allow you to verify your IP address, subnet mask, default gateway, and MAC address?

A. C:\>ipconfig/all

B. C:\>ping

C. C:\>ipconfig

D. C:\>ipstatus

Answer: A. C:\>ipconfig/all

Explanation: The ipconfig /all command in Microsoft Windows provides detailed information about the network configuration, including the IP address, subnet mask, default gateway, MAC address, and more.

26. Which bus is bidirectional?

A. Control bus

B. Data bus

C. Uni-bus

D. Address bus

Answer: B. Data bus

Explanation: The data bus is bidirectional, meaning it can carry data both to and from the processor, memory, and other peripherals. This allows the CPU to both send and receive data from other components.

27. Which one of the following is not an I/O command?

A. Status command

B. Output data command

C. Control command

D. Issue command

Answer: D. Issue command

Explanation: "Issue command" is not a standard term used in I/O operations. It is a general term and does not specifically relate to input/output (I/O) commands.

28. Which one of the following form attribute is used to assign the server side script file that processes the form data?

A. Action

B. Autocomplete

C. Method

D. Target

Answer: A. Action

Explanation: The action attribute in an HTML form specifies the URL of the server-side script or file that will process the form data when it is submitted. It defines where the form data should be sent for processing.

29. The first step in access control is ________

A. Audit

B. Authorization

C. Accreditation

D. Authentication

Answer: D. Authentication

Explanation: The first step in access control is authentication, which involves verifying the identity of a user or system attempting to access resources. Once the identity is confirmed, other processes like authorization (determining access permissions) and auditing (logging actions) follow.

30. Which class of languages is most suited to be represented by pushdown automata?

A. Context-free language

B. Regular language

C. Recursively enumerable language

D. Unrestricted language

Answer: A. Context-free language

Explanation: Pushdown automata (PDA) are most suited for recognizing context-free languages (CFL). PDAs are finite automata equipped with a stack, which allows them to handle a broader class of languages than finite automata alone, making them suitable for recognizing languages that require a form of memory, such as those generated by context-free grammars.

31. How are the objects organized in the HTML DOM?

A. Hierarchy

B. Queue

C. Stack

D. Class-wise

Answer: A. Hierarchy

Explanation: Objects in the HTML DOM (Document Object Model) are organized in a hierarchical tree structure. In this tree, each node represents an object, and the nodes are related to each other in a parent-child hierarchy. The root node is the <html> element, and other elements are nested within it as children, following the document's structure.

32. Which one of the following is the output of syntax analysis phase?

A. Syntax tree

B. Annotated parse tree

C. Abstract syntax tree

D. Pattern

Answer: C. Abstract syntax tree

Explanation: The output of the syntax analysis (or parsing) phase in a compiler is typically an Abstract Syntax Tree (AST). This tree represents the hierarchical syntactic structure of the source code, abstracting away certain details (such as parentheses or grammar-specific symbols) to focus on the structure of the program.

33. What does the if-else statement allow you to do in C++?

A. Declare and define functions

B. Allocate memory dynamically

C. Execute a block of code only if a condition is true, otherwise execute another block of code

D. Repeat a block of code multiple times

Answer: C. Execute a block of code only if a condition is true, otherwise execute another block of code

Explanation: In C++, the if-else statement allows conditional execution of code. If the condition in the if statement evaluates to true, the code within the if block is executed. If the condition is false, the code in the else block (if present) is executed instead.

34. One of the following is NOT TRUE about the inheritance?

A. The subclass can add variables, but cannot redefine them.

B. The derived class (subclass) is allowed to add methods or redefine them.

C. A feature that allow us to derive class from another class.

D. Allows programmers to customize with actually modifying the original class (the superclass).

Answer: A. The subclass can add variables, but cannot redefine them.

Explanation: This statement is NOT TRUE about inheritance. In fact, a subclass can both add new variables and redefine existing variables (though redefining variables can sometimes lead to issues with shadowing or overriding depending on the language). In languages like C++, the subclass can also override methods and, with careful use of access specifiers, control how inherited members are accessed.

35. Which one of the following is an efficient top-down parser?

A. Recursive-descent parser

B. Shift-reduce parser

C. Non-recursive predictive parser

D. LR parser

Answer: C. Non-recursive predictive parser

Explanation: A non-recursive predictive parser is an efficient top-down parser that uses a parsing table to decide which rule to apply at each step. It avoids the overhead of recursion and backtracking, making it more efficient than a recursive-descent parser in many cases.

36. Which objective of modern operating systems involves providing a consistent and convenient environment for running applications?

A. Maximizing hardware utilization

B. Ensuring fault tolerance

C. Providing security and protection

D. Providing a user-friendly interface

Answer: D. Providing a user-friendly interface

Explanation: One of the key objectives of modern operating systems is to provide a consistent and user-friendly interface that makes it convenient to run applications and manage resources. This helps users and developers interact with the system in a seamless manner without needing to understand the complexities of the underlying hardware.

37. Which one of the following IS NOT the concern of availability?

A. Fault Tolerance

B. Alteration of resources

C. Usefulness of resources

D. Fair allocation of resources

Answer: B. Alteration of resources

Explanation: Alteration of resources is related to security and integrity, not availability. It refers to unauthorized modification or tampering with resources, which compromises their integrity.

38. The simple strategy in which the root node is expanded first, then all the successors of the root node are expanded next, then their successors, and so on which is Expand the tree layer by layer, refers to:

A. Greedy best-first search

B. Depth-first search (DFS)

C. Depth limited search (DLS)

D. Breadth-first search (BFS)

Answer: D. Breadth-first search (BFS)

Explanation: In Breadth-first search (BFS), the root node is expanded first, followed by all of its direct successors (i.e., the nodes at the next level of the tree), and then the successors of those nodes, continuing level by level. This strategy ensures that the tree is expanded layer by layer.

39. Person A sends a message to person B, while the message is transmitted; person C has gained an access to the message. Which category of an attack is performed by person C?

A. Modification

B. Interception

C. Interruption

D. Fabrication

Answer: B. Interception

Explanation: When Person C gains access to a message during its transmission between Person A and Person B, it is considered an interception attack. This type of attack involves unauthorized access to information, which can compromise confidentiality.

40. Which data structure is used to store information about the source program constructs?

A. 2D-array

B. Stack

C. Tree

D. Queue

Answer: C. Tree

Explanation: A tree data structure, specifically an Abstract Syntax Tree (AST) or parse tree, is used to store information about the source program constructs. It represents the hierarchical structure of the program's syntax, as produced by the syntax analysis phase of a compiler. Each node in the tree represents a construct in the source code, such as expressions, statements, or declarations.

41. In which network topology consists of a single line cable called a backbone connecting all nodes on a network without intervening connected devices?

A. Bus

B. Ring

C. Star

D. Mesh

Answer: A. Bus

Explanation: In a bus topology, all nodes are connected to a single central cable called the backbone. Data sent by a node is transmitted along the backbone, and all other nodes receive the data, but only the intended recipient processes it. This topology is simple and inexpensive but can suffer from performance issues as more devices are added.

42. Which database design concept is commonly used in e-commerce platforms to efficiently store and retrieve customer order information?

A. Normalization

B. Denormalization

C. Partitioning

D. Indexing

Answer: B. Denormalization

Explanation: In e-commerce platforms, denormalization is commonly used to efficiently store and retrieve customer order information. Denormalization involves combining tables and reducing the need for joins by adding redundant data to speed up read-heavy operations, which is crucial for performance in real-time environments like e-commerce. It helps optimize database queries for faster access, especially when dealing with large volumes of data like customer orders.

43. Which one of the following terms refers to the techniques used for deciphering a cipher text without any knowledge of the enciphering mechanism involved?

A. Cryptography

B. Cryptology

C. Cryptoanalysis

D. Decryption

Answer: C. Cryptoanalysis

Explanation: Cryptoanalysis refers to the techniques used for deciphering ciphertext without prior knowledge of the encryption mechanism or key. It is the study and practice of breaking encryption systems and finding vulnerabilities in cryptographic algorithms.

44. All of the followings are scheduling criteria EXCEPT?

A. CPU utilization

B. Min Turnaround Time

C. Max Throughput

D. Max Response Time

Answer: D. Max Response Time

Explanation: In the context of scheduling criteria for operating systems, the goal is usually to minimize response time, not maximize it.

45. Given a binary search tree, which traversal type would print the values in the nodes in sorted order?

A. Pre-order

B. Pre-order and Post-order

C. In-order

D. Post-order

Answer: C. In-order

Explanation: In a binary search tree (BST), in-order traversal visits the nodes in a specific order: it first recursively visits the left subtree, then the current node, and finally the right subtree. This traversal will print the values in the nodes in sorted order because, in a BST, for any node, all values in the left subtree are smaller, and all values in the right subtree are larger.

46. The user who programs a computer in machine or assembly language must be aware of the following EXCEPT:

A. The register set

B. The type of data supported by the instructions

C. The analysis model

D. The memory structure

Answer: C. The analysis model

Explanation: The analysis model is not typically a concern for the programmer working in machine or assembly language. The analysis model is more relevant in the context of software design and higher-level program analysis rather than direct interaction with hardware and memory.

47. In computer programming, what is the purpose of control structures such as loops and conditional statements?

A. To control the flow of program execution

B. To perform mathematical operations

C. To define the data types used in a program

D. To allocate memory for variables

Answer: A. To control the flow of program execution

Explanation: Control structures like loops (e.g., for, while) and conditional statements (e.g., if, else) are used to manage the flow of program execution. They allow the program to make decisions (conditional statements) and repeat actions (loops), enabling more complex and dynamic behavior.

48. Which of the following IS NOT a component of Turing machine?

A. Read-write head

B. Control Unit

C. Stack

D. Tape

Answer: C. Stack

Explanation: A stack is not a component of a Turing machine. However, stacks are used in other computational models, such as in pushdown automata, which are different from Turing machines.

49. When encountering a syntax error in code, what is the final step you should take to resolve it?

A. Rewrite the entire

B. Ignore the error and continue execution

C. Use a debugger tool

D. Check the code for types or incorrect syntax

Answer: D. Check the code for types or incorrect syntax

Explanation: When encountering a syntax error in your code, the final step should involve carefully checking the code for types or incorrect syntax. This includes: Ensuring that all parentheses, brackets, and braces are properly matched, Verifying the correct use of keywords, operators, and punctuation and Checking for typos or misspellings in variable names, function names, or commands.

50. Which string from the following is member of the language represented by the regular expression a*(b+c)?

A. aab

B. bc

C. aaa

D. aaabc

Answer: A. aab

Explanation: The regular expression a(b+c)* can be broken down as follows:

a*: This means zero or more occurrences of the character a.

(b+c): This means either a b or a c.

51. When to use the Spiral model?

A. Where business modeling is clearly understood and in use.

B. When it is too risky to develop the overall system as one block.

C. When requirements are understood in the early stages but can change during the project's life cycle.

D. When requirements are unstable and constantly changing.

Answer: B. When it is too risky to develop the overall system as one block.

Explanation: The Spiral model is designed for projects where the risks of developing the entire system in one go are too high. It emphasizes iterative development, risk assessment, and refinement, making it ideal for complex or high-risk projects where requirements may evolve or uncertainties need to be managed carefully.

52. In C++, what does the cout object typically represent?

A. Standard input stream

B. Keyboard input buffer

C. Error output stream

D. Standard output stream

Answer: D. Standard output stream

Explanation: In C++, the cout object is part of the standard library and represents the standard output stream. It is used to display output to the console.

53. a → b, L state transition in a Turing Machine means,

A. read a, write b and move left

B. delete a, read b and move left

C. write a, read b and move left

D. replace b by a and move left

Answer: A. read a, write b and move left

Explanation: In a Turing Machine, a transition notation like "a → b, L" generally means that the machine reads symbol a on the tape, replaces it with b, and then moves the tape head to the left (L).

54. What is a characteristic feature of round-robin scheduling in both preemptive and non-preemptive forms?

A. It assigns priorities to tasks based on their relative importance.

B. It always selects the task with the shortest remaining time.

C. It favors short tasks over long tasks.

D. It guarantees fairness by providing each task with an equal time slice.

Answer: D. It guarantees fairness by providing each task with an equal time slice.

Explanation: Round-robin scheduling is characterized by allocating each task a fixed time slice (or quantum) in a cyclic order, ensuring that all tasks receive an equal opportunity to execute. This approach promotes fairness and is widely used in time-sharing systems.

55. Indexes in database tables speed up the _______ operations.

A. Read operations

B. Delete operations

C. Update operations

D. Write operations

Answer: A. Read operations

Explanation: Indexes in database tables are primarily used to speed up read operations, such as searching and retrieving data. By creating an index on a column or set of columns, the database can quickly locate rows that match the query criteria, improving the performance of read queries. However, indexes can slightly slow down write operations (like inserts, updates, and deletes) due to the need to maintain the index structure.

56. What would be the asymptotic time complexity to add a node at the end of singly linked list, if the pointer is initially pointing to the head of the list?

A. O(n)

B. $\vartheta(1)$

C. $\vartheta(n)$

D. O(1)

Answer: A. O(n)

Explanation: In a singly linked list, if the pointer is initially pointing to the head, you need to traverse the entire list to reach the last node in order to add a new node at the end. This traversal requires nnn steps, where nnn is the number of nodes in the list, resulting in a time complexity of O(n) or $\vartheta(n)$.

57. Which learning technique in AI involves the use of labeled examples to train a model to make predictions or decisions?

A. Semi-supervised learning

B. Unsupervised learning

C. Supervised learning

D. Reinforcement learning

Answer: C. Supervised learning

Explanation: In supervised learning, a model is trained using labeled examples, where each input data point is paired with the correct output. This technique helps the model learn to make predictions or decisions based on the labeled training data.

58. A PC obtains its IP address from a DHCP server. If the PC is taken off the network for repair, what happens to the IP address configuration?

A. The address is returned to the pool for reuse when the lease expires.

B. The address lease is automatically renewed until the PC is returned.

C. The configuration is held by the server to be reissued when the PC is returned.

D. The configuration is permanent and nothing changes.

Answer: A. The address is returned to the pool for reuse when the lease expires.

Explanation: In DHCP (Dynamic Host Configuration Protocol), IP addresses are typically leased for a set period. If a device is disconnected from the network and its lease expires, the IP address is returned to the pool, making it available for other devices on the network. When the device reconnects, it may receive a new IP address, depending on availability.

59. The elements must be in sorted order in case of:

A. Quick search

B. Selection search

C. Binary search

D. Linear search

Answer: C. Binary search

Explanation: Binary search requires the elements to be in sorted order to function correctly. This search algorithm divides the search interval in half repeatedly, making it efficient for sorted arrays or lists. Without sorting, binary search cannot reliably locate elements.

60. Which analysis case calculates the upper bound on running time of an algorithm?

A. Optimal case

B. Best case

C. Average case

D. Worst case

Answer: D. Worst case

Explanation: The worst case analysis calculates the upper bound on the running time of an algorithm, representing the maximum time required to complete the algorithm for the most unfavorable input. It helps in understanding the worst performance scenario of an algorithm.

61. Which of the following methods can be used to refresh a webpage?

A. location.relocate

B. window.relocate

C. window.refresh

D. location.reload

Answer: D. location.reload

Explanation: In JavaScript, the location.reload() method is used to refresh or reload the current webpage. This can be useful to re-fetch the page from the server or reload the content without manually refreshing the browser.

62. If a sequence of operations: push(1), push(2), pop, push(1), push(2), pop, pop, pop, push(2), pop are performed on a stack, the sequence of popped out values are:

A. 2,1,2,2,1

B. 2,2,1,1,2

C. 2,1,2,2,2

D. 2,2,1,2,2

Answer: B. 2,2,1,1,2

63. Which software engineering principles are used to help reduce the programming complexity?

A. Structured and unstructured

B. Object Oriented Design

C. Abstraction and Decomposition

D. Functional and Non-functional

Answer: C. Abstraction and Decomposition

Explanation: Abstraction involves hiding the complex details of a system and providing a simplified interface, making it easier to understand and interact with. Decomposition breaks down a large, complex problem into smaller, more manageable parts, each of which can be solved independently, thereby simplifying the overall system.

64. In a grammar, if the production $S \rightarrow \lambda$ where S is a non-terminal is existed, such production is called:

A. Null production

B. Unit production

C. Secondary production

D. Sigma production

Answer: A. Null production

Explanation: A null production is a production rule in a grammar where a non-terminal symbol (in this case, S) produces the empty string (denoted as λ or ε). This type of production indicates that the non-terminal can be replaced by nothing, effectively "disappearing" from the string.

65. Recursion is a method in which the solution of a problem depends on:

A. Larger instances of the same problem

B. Smaller instances of the same problem

C. Larger instances of different problems

D. Smaller instances of different problems

Answer: B. Smaller instances of the same problem

Explanation: Recursion is a method in which a problem is solved by breaking it down into smaller instances of the same problem. The solution to the original problem is built upon the solutions to these smaller subproblems, typically by calling the same function repeatedly with smaller inputs until a base case is reached.

66. Which of the following is NOT TRUE about reasoning?

A. Knowledge system to do something.

B. A process of thinking, logically arguing, and drawing inference.

C. Theory of blind searching.

D. The act of deriving a conclusion from certain premises using a given methodology

Answer: C. Theory of blind searching.

Explanation: "Blind searching" refers to a search strategy in algorithms, particularly in artificial intelligence, where there is no information about the problem space (e.g., depth-first search). It is not directly related to reasoning, which typically involves logical thinking, drawing inferences, or deriving conclusions from known information.

67. In asymmetric cryptography, the message is decrypted using ______

A. Receiver's public key

B. Receiver's private key

C. Sender's private key

D. Sender's public key

Answer: B. Receiver's private key

Explanation: In asymmetric cryptography, also known as public-key cryptography, encryption and decryption use different keys. A message encrypted with the receiver's public key can only be decrypted using the receiver's private key. This ensures that only the receiver can decrypt the message, providing confidentiality.

68. When can we say two grammars are equivalent?

A. When the language derived from one of the grammars become the subset of the other

B. When the alphabets of the two grammars are the same

C. When the languages represented by both grammars are the same

D. When a single string is member of the language represented by both grammars

Answer: C. When the languages represented by both grammars are the same

Explanation: Two grammars are considered equivalent if they generate the same language—meaning both grammars can produce exactly the same set of strings, regardless of how the rules or symbols differ between the grammars. This is the key characteristic of equivalence between two grammars.

69. Which one of the following IS NOT a component of communication?

A. Distance

B. Source

C. Destination

D. Medium

Answer: A. Distance

Explanation: Distance is not considered a direct component of communication, although it may affect the medium or transmission of the message.

70. Which one of the following IS NOT a static check?

A. Type checking for compatibility

B. Uniqueness checks like redefined variables

C. A break instruction that is not in inclosing statement

D. Memory access errors

Answer: D. Memory access errors

Explanation: Memory access errors (such as accessing invalid memory or dereferencing null pointers) typically occur at runtime, not during static checking. Static checks are performed before the program is run, during compilation, or in the analysis phase.

71. ______ is a system that monitors traffic into and out of a network and automatically alerts personnel when suspicious traffic patterns occur, indicating a possible unauthorized intrusion attempt.

A. Anti-virus Software

B. IDS

C. Firewall

D. Router

Answer: B. IDS

Explanation: An Intrusion Detection System (IDS) monitors traffic into and out of a network, analyzing it for suspicious patterns or activities that could indicate unauthorized access or potential threats. When such activity is detected, the IDS generates alerts to notify administrators.

72. Which one of the following terms refers to the machine on which the requested resource is stored?

A. Port

B. Query

C. Host

D. Protocol

Answer: C. Host

Explanation: A host refers to the machine (server or device) on which the requested resource is stored. In a network context, the host is identified by its IP address or hostname, which allows clients to access resources like web pages, files, or applications.

73. Why heuristic (A*) search method?

A. To speed up search and edge lengths to compute optimal paths.

B. To find the path with the maximum distance from the start node to the goal node.

C. To find the path with the non-optimal searches from node to the goal node.

D. To find the path with the maximum cost from the start node to the goal node.

Answer: A. To speed up search and edge lengths to compute optimal paths.

Explanation: The A* (A-star) search method is a heuristic search algorithm that is widely used for finding the shortest path in graphs. It combines the cost to reach a node (g(n)) with a heuristic estimate of the cost to reach the goal from that node (h(n)).

74. When is the appropriate time to shut down a system?

A. Using regular schedules

B. For maintenance and upgrade

C. To check for security alerts

D. When there is a failure

Answer: B. For maintenance and upgrade

Explanation: Shutting down a system is appropriate when it is necessary for maintenance and upgrades, such as applying patches, upgrading hardware or software, or performing preventive maintenance. This ensures the system operates efficiently and securely after the updates.

75. Out of the following life cycle models which one can be considered as the most general model, and the others as specialization of it?

A. Spiral model

B. Prototyping model

C. Classical Waterfall model

D. Iterative Waterfall model

Answer: A. Spiral model

Explanation: The Spiral model can be considered the most general software development life cycle model. It is a risk-driven process that combines elements of iterative development and the systematic aspects of the Waterfall model. Other models, such as the Waterfall, Iterative, and Prototyping models, can be seen as specializations or simplified versions of the Spiral model, depending on the project's needs and focus areas.

76. Which one of the following cryptography systems uses the same key for encryption and decryption?

A. Public key cryptography

B. Parallel cryptography

C. Asymmetric cryptography

D. Symmetric cryptography

Answer: D. Symmetric cryptography

Explanation: In symmetric cryptography, the same key is used for both encryption and decryption of data. It is efficient and commonly used for encrypting large amounts of data but requires secure sharing of the key between parties.

77. In which network topology should every computer have N-1 I/O ports when there is N number of computers in the network?

A. Star

B. Mesh

C. Bus

D. Ring

Answer: B. Mesh

Explanation: In a mesh topology, every computer is directly connected to every other computer in the network. If there are N computers, each computer requires N-1 I/O ports for direct connections to the other N-1 computers. This topology provides high redundancy and reliability but comes at the cost of high hardware and cabling requirements.

78. Which C++ statement is used to terminate a loop prematurely and transfer control to the next iteration of the loop?

A. break

B. exit

C. return

D. continue

Answer: D. continue

Explanation: In C++, the continue statement is used to prematurely terminate the current iteration of a loop and immediately transfer control to the beginning of the next iteration. It skips any remaining statements in the loop body for the current iteration.

79. Which one of the following disk interfaces is used as a standard interface for PCs?

A. SCSI

B. iSCSI

C. IDE

D. USB

Answer: C. IDE

Explanation: IDE, also known as ATA (Advanced Technology Attachment), has been widely used as a standard interface for connecting storage devices (like hard drives and CD/DVD drives) in PCs. It integrates the drive controller directly into the drive, simplifying the connection process and reducing costs.

80. In a system susceptible to deadlock, what does the “no preemption” condition imply?

A. Resources can be preempted and assigned to other processes.

B. Processes cannot request additional resources while holding others.

C. Resources cannot be preempted while they are in use.

D. Processes must release all resources when blocked.

Answer: C. Resources cannot be preempted while they are in use.

Explanation: The "no preemption" condition in the context of deadlock means that once a process has acquired a resource, it cannot be forcibly taken away (preempted) from that process until it has completed its task and released the resource voluntarily. This can lead to deadlock in situations where processes hold

resources and wait for others, but no resource can be taken away to break the cycle.

81. In a columnar database, data is stored:

A. Row-wise

B. In a hierarchical structure

C. Column-wise

D. In a network model

Answer: C. Column-wise

Explanation: In a columnar database, data is stored column-wise rather than row-wise. This means that all the values for a single column are stored together, making it particularly efficient for querying large datasets, especially when only a few columns are needed for a query. It is commonly used in analytical databases and data warehouses where operations often require reading specific columns rather than entire rows.

82. In computer networking, _____ refers to how various nodes, devices, and connections on your network are physically or logically arranged in relation to each other.

A. Network layer

B. Network installation

C. Network protocol

D. Network topology

Answer: D. Network topology

Explanation: Network topology refers to the physical or logical arrangement of various nodes (devices) and the connections (links) between them within a network. It defines how devices communicate and how data flows across the network. Common types of network topologies include star, bus, ring, mesh, and hybrid topologies.

83. A module is said to have logical cohesion, if:

A. All the functions of the module are executed within the same time span.

B. All elements of the module depend on their interface complexity.

C. It performs a set of tasks that relate to each other very loosely.

D. All elements of the module perform similar operations.

Answer: D. All elements of the module perform similar operations.

Explanation: Logical cohesion refers to a situation where the elements of a module perform operations that are related logically, even though they might not be functionally related. In other words, a module with logical cohesion might include different tasks that are grouped together because they share a similar logical context.

84. Which of the following IS NOT a type of database management system (DBMS)?

A. PostgreSQL

B. MongoDB

C. Java

D. MySQL

Answer: C. Java

Explanation: Java is a programming language and not a type of database management system (DBMS). It can be used to interact with databases but is not itself a DBMS.

85. If queue is implemented using arrays, what would be the worst run time complexity of queue and dequeue operations?

A. O(n), O(1)

B. O(1), O(n)

C. O(1), O(1)

D. O(n), O(n)

Answer: B. O(1), O(n)

Explanation: When a queue is implemented using an array, the enqueue (insert) operation can generally be performed in O(1) time because the new element is simply added at the end of the array. However, the dequeue (remove) operation can take O(n) time in the worst case when the array is implemented in such a way that after removing an element.

86. Which tool is used to collect codes separately compiled or assembled in different object files into a file that directly executable?

A. Linker

B. Loader

C. Assembler

D. Resolver

Answer: A. Linker

Explanation: A linker is a tool used to combine object files (which are separately compiled or assembled code files) into a single executable file. It resolves references between different object files, such as function calls and variable references, and combines the code and data into a unified program.

87. Which one of the following IS NOT a characteristic of an algorithm?

A. Definiteness

B. Output

C. Finiteness

D. Efficiency

Answer: D. Efficiency

Explanation: While efficiency is an important consideration when analyzing algorithms (in terms of time and space complexity), it is not a fundamental characteristic that defines an algorithm.

88. One of the following IS NOT characteristics of Java?

A. Java is Object-Oriented

B. Java is Platform Independent

C. Java is difficult to learn

D. Java is architecture neutral

Answer: C. Java is difficult to learn

Explanation: Java is generally considered an easy-to-learn language, especially for those with prior programming experience. It was designed with simplicity in mind, and its syntax is largely based on C, which makes it more approachable.

89. In data communication, data rate refers how fast we can send data and is measured in bits per second. Which one of the following is not a factor for data rate?

A. Travel distance

B. Bandwidth available

C. Level of noise

D. Levels of the signals

Answer: A. Travel distance

Explanation: In data communication, travel distance is not a direct factor affecting the data rate (how fast data can be sent).

90. Which layer of the OSI MODEL ensures that the whole message (group of packets) arrives to the destination intact and in order?

A. Session layer

B. Transport layer

C. Data link layer

D. Network layer

Answer: B. Transport layer

Explanation: The Transport layer (Layer 4) of the OSI model ensures reliable data transfer between two devices. It guarantees that the entire message (or group of packets) is delivered to the destination intact and in the correct order. It handles error detection and correction, flow control, and retransmission of lost packets. Protocols such as TCP (Transmission Control Protocol) operate at this layer, providing these guarantees.

91. Which system requirement specification includes human computer interface issues from the system?

A. Functional requirements of the system

B. Non-functional requirements of the system

C. Goals of implementation

D. System Testing

Answer: B. Non-functional requirements of the system

Explanation: Non-functional requirements describe the qualities and characteristics that the system should exhibit, such as performance, usability, security, and human-computer interaction (HCI) issues. These requirements focus on how the system performs rather than what it does.

92. In a classical waterfall model, which phase precedes the design phase?

A. Requirements analysis and specification

B. Maintenance

C. Feasibility study

D. Coding and unit testing

Answer: A. Requirements analysis and specification

Explanation: In the classical waterfall model, the Requirements analysis and specification phase precedes the Design phase. During the requirements phase, the system's requirements are gathered and documented. This information is then used to design the system in the next phase.

93. In MySQL architecture, the function of the ______ is to keep copies of data for retrieval later, in case of a loss of data.

A. Buffer Manager

B. Transaction Manager

C. Query Engine

D. Recovery Manager

Answer: D. Recovery Manager

Explanation: In MySQL architecture, the Recovery Manager is responsible for maintaining copies of data (through logs and backups) so that the system can recover in case of data loss or failure. It ensures that after a crash or failure, the database can be restored to a consistent state by using the stored recovery information, such as the transaction logs.

94. What role does machine learning play in understanding human perception?

A. Machine learning helps in simulating human perception through neural networks.

B. Machine learning has no relevance to understanding human perception.

C. Machine learning is not capable of analyzing perceptual data.

D. Machine learning only focuses on logical reasoning and decision-making.

Answer: A. Machine learning helps in simulating human perception through neural networks.

Explanation: Machine learning, especially through techniques like neural networks, is used to model and simulate aspects of human perception. For example, deep learning models, which are inspired by the structure and functioning of the human brain, can analyze sensory data (such as images, sounds, or text) to mimic human-like perception.

95. The amount of memory needs to run to completion is known as:

A. Space complexity

B. Time complexity

C. Worst case

D. Best case

Answer: A. Space complexity

Explanation: Space complexity refers to the amount of memory required by an algorithm to run to completion, as a function of the size of the input. It includes both the memory required for storing input data and any extra memory used by the algorithm for processing and temporary storage.

96. Which of the following IS NOT a type of database model?

A. Linear

B. Object-Oriented

C. Hierarchical

D. Relational

Answer: A. Linear

Explanation: Linear is not a recognized type of database model.

THANK YOU!!!